ETH zürich

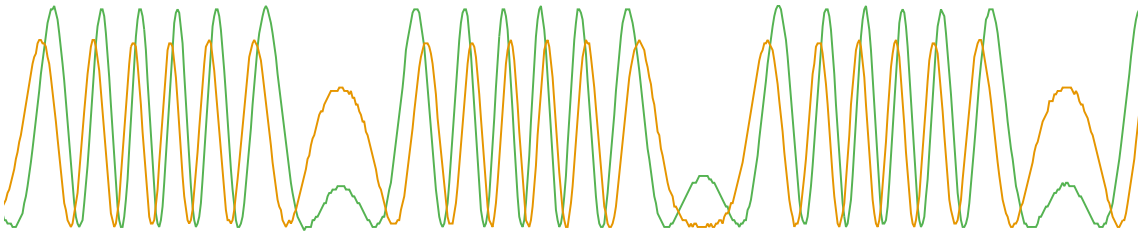
Department logo

Department of Everything
Institute of Nothing

# A Very Inspiring Title

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Bachelor Thesis

12 November 2026

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A

# Abstract

This example shows the smallest useful configuration for a thesis. The document body follows the configuration below, while the lists and bibliography are added afterwards.

# B

# Contents

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# 1

# Introduction

This is the document body. Write chapters directly in this file, or split them into separate files with `#include("chapters/introduction.typ")`.

The modular thesis template (MTT) provides reasonable defaults while leaving the document structure in your control. The tables below summarize the functions provided by the template. The examples in the following chapters demonstrate some of them in more detail.

## 1.1 The main template function

The main entry point configures the document and assembles its parts.

simple.typ (from README) typst

```typst
#import "@preview/mtt-thesis:0.1.0": *

#show: mtt-thesis.with(
  coverpage: mtt-coverpage(
    org-units: ("Department of Everything",),
    title: [A Very Inspiring Title],
    project-type: "Bachelor Thesis",
    authors: ((name: "Jane Doe", email: "janedoe@university.org"),),
  ),
  before: (
    mtt-abstract[This is the abstract.],
    mtt-toc(),
  ),
  after: (
    mtt-figures(),
    mtt-tables(),
    mtt-bibliography(
      bibliography("references.bib"), // requires this file existing
    ),
  ),
)

= Introduction

Your thesis starts here.
```

Code Sample 1: A simple example of `#mtt-thesis`.

## 1.2 Functions that generate pages

These functions create complete pages or page-sized sections. They are mainly intended as appendix modules, as illustrated in Code Sample 1 on lines 10-20.

| Function | Purpose |
|---|---|
| mtt-coverpage | Create the thesis cover page from institutional, author, and project metadata. |
| mtt-abstract | Create a page containing a titled abstract. |
| mtt-toc | Create a formatted table of contents page. |
| mtt-figures | Create a list of figures of a selected figure kind. |
| mtt-tables | Create a list containing only table figures. |
| mtt-codes | Create a list containing only code sample figures. |
| mtt-todos | Create a list containing only todo figures. |
| mtt-acronyms | Create an acronym index with configurable sorting and layout. |
| mtt-bibliography | Create a titled bibliography page. |
| mtt-content-frame | Place content in a centered, titled frame. |
| mtt-typst-guide | Add the bundled Typst guide as a standalone page. |

Table 1: Functions that generate pages.

## 1.3 Smaller helper functions

These functions support common content patterns without creating a complete document page.

| Function | Purpose |
|---|---|
| mtt-placeholder | Display a task-style placeholder with an explanation and usage snippet. |
| mtt-breakable-figure | Place content in a breakable figure and infer its figure kind when requested. |
| mtt-code-sample | Place a breakable code sample in a raw figure with a configurable caption and label. |
| mtt-space-equally | Basically a grid, but with equal spacing in between the elements. Useful to distribute elements along a line or page. |

Table 2: Smaller helper functions.

2

# Code of this File

This section reproduces the source code used to generate this entire PDF.

example.typ typst

```typst
// A small, complete thesis built with MTT.
// Copy this file into your project and replace the example content. The

#import "@preview/mtt-thesis:0.1.0": *

// Acronyms are optional. Define them before using mtt-acronyms() below.
#import "@preview/acrostiche:0.7.0": *
#init-acronyms((
  "MTT": ("modular thesis template", "modular thesis templates"),
  "API": ("application programming interface", "application programming interfaces"),
))

#import "@preview/codly:2.0.0" as codly
#show raw.where(block: true): codly.new
#show: codly.theme("one-light")


// mtt-thesis controls the document-wide layout. The order of items in
// `before` and `after` is the order in which their pages are rendered.
#show: mtt-thesis.with(
  coverpage: mtt-coverpage(
    logo-left: image("template/eth_logo_kurz_cropped.svg", height: 25pt),
    logo-center: [],
    logo-right: rect(height: 35pt)[Department logo],
    org-units: (
      "Department of Everything",
      "Institute of Nothing",
    ),
    title: [A Very Inspiring Title],
    project-type: "Bachelor Thesis",
    date: datetime.today(),
    cover-image: image("template/cover-image.svg", width: 100%),
    authors: (
```

```
      (name: "Jane Doe", email: "janedoe@university.org"),
      (name: "John Doe", email: "johndoe@university.org"),
    ),
    authors-horizontal: true,
    advisors: ((name: "Advisor One", email: "advisor.one@university.org"),),
    professors: ((name: "Prof. Dr. Professor", email: "professor@university.org"),),
    empty-page: false,
  ),
  before: (
    mtt-abstract[
      This example shows the smallest useful configuration for a thesis. The
      document body follows the configuration below, while the lists and
      bibliography are added afterwards.
    ],
    mtt-toc(),
  ),
  after: (
    mtt-figures(),
    mtt-tables(),
    mtt-codes(),
    mtt-acronyms(),
    mtt-bibliography(
      bibliography(
        "references.bib",
        style: "american-physics-society",
        full: true,
        title: none,
      ),
    ),
    // Add this optional guide when starting to learn Typst.
    mtt-typst-guide(),
  ),
)

= Introduction

This is the document body. Write chapters directly in this file, or split them into separate files with `#include("chapters/introduction.typ")`.

The #acr("MTT") provides reasonable defaults while leaving the document structure in your control. The tables below summarize the functions provided by the template. The examples in the following chapters demonstrate some of them
```

```
in more detail.

== The main template function

The main entry point configures the document and assembles its parts.

#mtt-code-sample(
  caption: [A simple example of `#mtt-thesis`.],
  ref: "code:simple"
)[
  #codly.new(
    raw(read("README.md").split(regex("```(typ\n)?")).at(1), lang:
    ↪ "typst", block: true),
    file: "simple.typ (from README)",
    indent-guides: (width: 4),
    rainbow: true,
    wrap-marker: [↪],
    breakable: true,
  )
]


== Functions that generate pages

These functions create complete pages or page-sized sections. They are
↪ mainly intended as appendix modules, as illustrated in @code:simple on
↪ lines 10-20.

#figure(
  table(
    columns: (auto, 1fr),
    table.header([*Function*], [*Purpose*]),
    [mtt-coverpage],
    [Create the thesis cover page from institutional, author, and project
    ↪ metadata.],
    [mtt-abstract],
    [Create a page containing a titled abstract.],
    [mtt-toc],
    [Create a formatted table of contents page.],
    [mtt-figures],
    [Create a list of figures of a selected figure kind.],
    [mtt-tables],
    [Create a list containing only table figures.],
    [mtt-codes],
    [Create a list containing only code sample figures.],
    [mtt-todos],
```

```
    [Create a list containing only todo figures.],
    [mtt-acronyms],
    [Create an acronym index with configurable sorting and layout.],
    [mtt-bibliography],
    [Create a titled bibliography page.],
    [mtt-content-frame],
    [Place content in a centered, titled frame.],
    [mtt-typst-guide],
    [Add the bundled Typst guide as a standalone page.],
  ),
  caption: [Functions that generate pages.],
)

== Smaller helper functions

These functions support common content patterns without creating a complete
document page.

#figure(
  table(
    columns: (auto, 1fr),
    table.header([*Function*], [*Purpose*]),
    [mtt-placeholder],
    [Display a task-style placeholder with an explanation and usage snippet.],
    [mtt-breakable-figure],
    [Place content in a breakable figure and infer its figure kind when requested.],
    [mtt-code-sample],
    [Place a breakable code sample in a raw figure with a configurable caption and label.],
    [mtt-space-equally],
    [Basically a grid, but with equal spacing in between the elements. \
    Useful to distribute elements along a line or page.],
  ),
  caption: [Smaller helper functions.],
)


= Code of this File <ch:code>

This section reproduces the source code used to generate this entire PDF.

#mtt-code-sample(
  caption: [The Typst code generating this file.],
  ref: "code:example.typ"
```

```
)[
  #codly.new(
    raw(read("example.typ"), lang: "typst", block: true),
    file: "example.typ",
    indent-guides: (width: 4),
    rainbow: true,
    wrap-marker: [↪],
    breakable: true,
  )
]

```

Code Sample 2: The Typst code generating this file.

C

# List of Figures

D

# List of Tables

Table 1 Functions that generate pages. 2

Table 2 Smaller helper functions. 2

E

# List of Code Samples

Code Sample 1 A simple example of `#mtt-thesis`. 1

Code Sample 2 The Typst code generating this file. 7

# F

# List of Acronyms

**API** application programming interface

**MTT** modular thesis template

G

# Bibliography

[1] P. Virtanen et al., SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python, Nature Methods **17**, 261 (2020).

# H

# Typst Quick Guide

Typst is a modern typesetting system designed as a faster, more ergonomic alternative to LaTeX. This guide covers the most common tasks for writing a thesis, with LaTeX equivalents shown alongside every Typst example.

> **Info**
>
> The full Typst documentation is available at https://typst.app/docs. The interactive web app at https://typst.app lets you experiment without any local installation.

## H.1 Text Formatting

| Feature | LaTeX | Typst |
|---|---|---|
| Bold | `\textbf{text}` | `*text*` or `#strong[text]` |
| Italic | `\textit{text}` | `_text_` or `#emph[text]` |
| Monospace | `\texttt{text}` | `` `text` `` |
| Superscript | `x\textsuperscript{2}` | `x#super[2]` |
| Subscript | `x\textsubscript{i}` | `x#sub[i]` |
| Forced linebreak | `\\` | `\` (backslash + space) |
| Non-breaking space | `~` | `~` |
| Quotation marks | ``` ``text'' ``` | `"text"` (auto smart quotes) |
| Hyperlink | `\href{url}{label}` | `#link("url")[label]` |

## H.2 Mathematics

Typst uses `$...$` for both inline and display math. Adding spaces *inside* the delimiters switches to display (block) mode:

| Feature | LaTeX | Typst |
|---|---|---|
| Inline math | `$x^2$` | `$x^2$` |
| Display math | `\[ x^2 \]` | `$ x^2 $` (spaces inside `$`) |
| Fraction | `\frac{a}{b}` | `$a/b$` |
| Square root | `\sqrt{x}` | `$sqrt(x)$` |
| Sum / integral | `\sum_{i=0}^{n}` | `$sum_(i=0)^n$` |
| Greek letters | `\alpha, \beta` | `$alpha, beta$` |
| Bold math | `\mathbf{A}` | `$bold(A)$` |
| Aligned equations | `\begin{align}..\end{align}` | `$ a &= b \ c &= d $` |

> **Tip**
>
> Math function names in Typst have no backslash: `sin`, `cos`, `lim`, `max`, etc. See the symbol reference for the full list of available symbols.

## H.3 Figures

**Code**

```typ
#figure(
  image("figures/arch.svg", width: 80%),
  caption: [A descriptive caption.],
) <fig:arch>

// Reference it anywhere in the text:
As shown in @fig:arch ...
```

Labels are placed with `<label>` directly after the closing `)` of `figure()`. `@label` is the equivalent of `\autoref{}` — Typst automatically prefixes "Figure", "Table", "Section", etc.

## H.4 Tables

Typst tables use `stroke: none` with `table.hline` for booktabs-style rules:

**Code**

```typ
#figure(
  table(
    columns: (auto, 1fr, 1fr),
    stroke: none,
    table.hline(stroke: 1.5pt),               // \toprule
    table.header([*Config*], [*Freq.*], [*Area*]),
    table.hline(stroke: 0.75pt),              // \midrule
    [Baseline], [500 MHz], [142 500 µm²],
    [Opt.],     [700 MHz], [164 900 µm²],
    table.hline(stroke: 1.5pt),               // \bottomrule
  ),
  caption: [Synthesis results.],
) <tab:results>
```

Column widths accept `auto` (fit content), fixed lengths (`3cm`), or fractional (`1fr`, `2fr`) to share available space proportionally.

## H.5 Labels and Cross-References

Code

```typ
= Introduction <chap:intro>     // label a heading
#figure(...) <fig:arch>         // label a figure

As shown in @fig:arch and discussed in @chap:intro ...
```

## H.6 Bibliography and Citations

Code

```typ
#show: thesis.with(
  bibliography: bibliography("references.bib", style: "ieee", full: true),
)

As shown in @smith2023 ...      // equivalent to \cite{smith2023}
```

The `.bib` format is identical to LaTeX. The bibliography is rendered automatically at the end of the document.

## H.7 Lists

Code

```typ
// Unordered list (\begin{itemize})
- First item
- Second item
  - Nested item

// Ordered list (\begin{enumerate})
+ First item
+ Second item

// Term list (\begin{description})
/ Term: Definition of the term.
```

## H.8 Including External Files

**Code**

```typ
// Render content from another file (\input equivalent):
include "/chapters/introduction.typ"

// Import bindings without rendering (for macros / dicts):
#import "/acronyms.typ": acronyms
```

## H.9 Useful Packages

| Package | Purpose |
|---|---|
| `@preview/cetz` | Drawing and diagrams (TikZ equivalent) |
| `@preview/acrostiche` | Acronym management (`\newacronym` equivalent) |
| `@preview/gentle-clues` | Callout / admonition boxes |
| `@preview/lovelace` | Algorithm pseudocode |
| `@preview/equate` | Numbered sub-equations |
| `@preview/dashy-todo` | Tracking to-dos |

Packages are imported with a version-pinned line — Typst downloads them automatically on first use, no manual installation needed:

**Code**

```typ
#import "@preview/cetz:0.4.2": canvas, draw
```

Browse the full package registry at https://typst.app/universe.